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“Real-time model-based control of a water tank system”

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Abstract

This report focuses on the design, development, and implementation of a real-time control system specifically crafted for a water tank experimental setup. The system is powered by an Arduino Uno R3 microcontroller, with an ultrasonic sensor providing precise water level measurements and a National Instruments (NI) USB 6008 module facilitating accurate data collection. The system's primary function is to maintain a target water level by adjusting the inflow rate through a Proportional-Integral-Derivative (PID) controller, which operates based on real-time feedback.

The project development involved running extensive simulations in MATLAB/Simulink to fine-tune the PID controller parameters, followed by thorough experimental validation using LabVIEW to evaluate real-world performance. Results from these simulations and practical tests demonstrate that the system effectively maintains accurate water levels, with minimal overshoot, quick settling times, and strong stability. These outcomes suggest that the control system could be effectively applied in various industrial settings for fluid management and process control.

Contents

Abstract 1

1 Introduction 5
- 1.1 Background 5
- 1.2 Problem Statement 6
- 1.3 Objectives 6
- 1.4 Scope and Limitations 6

2 Literature Review 7
- 2.1 Integration of Microcontrollers and Sensors 7
- 2.2 Arduino and NI USB 6008 in Real-Time Applications 7
- 2.3 Model-Based Control 7
- 2.4 Advances in Fluid Control Systems 8

3 System Design and Development 8
- 3.1 Overview of System Architecture 8
- 3.2 Hardware Configuration 8
- 3.3 Software Development 9
- 3.4 Mathematical Modeling 9

4 Experimental Setup 10
- 4.1 Water Tank Configuration 11
- 4.2 Electric Motor and Pumping Mechanism 12
- 4.3 Implementation and Integration 12
- 4.4 Ultrasonic Sensor and Arduino Integration 12
- 4.5 National Instrument NI 6008 Board and Simulink Integration 13
- 4.6 Software Configuration and Calibration 15

5 Methodology 15
- 5.1 System Identification 15
- 5.2 Controller Design 16
- 5.3 Implementation and Integration 16
- 5.4 Testing and Validation 17

6 Analysis 17
- 6.1 Data Analysis 17
- 6.2 Experimental Results 17
- 6.3 Comparison with Simulated Models 18
- 6.4 Performance Metrics 18
- 6.5 Challenges and Solutions 18

7 Ultrasonic Sensor HC-SR04 and Arduino code 19
- 7.1 Operating Principle 19
- 7.2 Displaying Distance Measurements Using an Ultrasonic Sensor and LCD with Arduino 20
- 7.3 Components Overview 21
- 7.4 Application Insights 22
- 7.5 HC-SR04 Ultrasonic Sensor: Dimensions and 3D Model Overview 22
- 7.6 3D Model Details 23
- 7.7 Benefits of the 3D Model 23

7.8 Arduino Code 23

8 Future Work 24

8.1 Summary of Findings 24

8.2 Contributions to the Field 24

8.3 Future Enhancements 24

9 Graphs, Data Tables, and Analysis 24

9.1 Graphs 24

9.2 System Response Graphs 25

9.3 Detailed Data Tables 26

9.4 In-Depth Analysis 26

10 Results 28

10.1 Water Level Measurement and Control Accuracy: 29

10.2 Response Time and System Dynamics: 30

10.3 Visualization and Data Logging 31

10.4 Impact of Sensor Placement and Calibration 31

10.5 System Limitations and Observations 32

11 Discussion 32

11.1 Performance of the Control System 32

11.2 Comparison with Existing Systems 32

11.3 Potential for Future Improvements 33

11.4 Real-World Applications 33

12 Conclusion 34

13 References 35

Figure 1 Water Tank System 10
Figure 2 Water Tank 11
Figure 3 Motor and Starter 12
Figure 4 Arduino connection 12
Figure 5 Ultrasonic sensor 13
Figure 6 National Instrument 13
Figure 7 Isometric view Float plate 14
Figure 8 Top view Float pipe 15
Figure 9 Ultrasonic sensor Distance 18
Figure 10 Arduino to sensor connection 19
Figure 11 Display 21
Figure 12 Display Connector 21
Figure 13 Ultrasonic sensor 22
Figure 14 LabVIEW Graph 29
Figure 15 Arduino graph 29
Figure 16 water tank measurement 29
Figure 17 LabVIEW output 31
Figure 18 LabVIEW blocks 31

Table 1 Sensor Specification 20
Table 2 LabVIEW data 26
Table 3 Sensor data 26

1 Introduction

The 1980s and 1990s introduced model-based control techniques that significantly improved the precision of water tank management. These strategies utilize mathematical models to forecast future system states and adjust control actions based on these predictions. Model Predictive Control (MPC) is one such technique that has gained extensive attention for its application in water-tank systems. MPC is particularly effective for handling complex systems due to its capability to manage multiple inputs, outputs, constraints, and nonlinearities. Research by Maciejowski (2002) and Mayne et al. (2000) has proven MPC's effectiveness in various fluid management scenarios.

In both industrial and research settings, precise control of fluid levels is essential. The development of real-time control systems for water tanks aims to continuously monitor and adjust to maintain the target water level. This project was conducted in a hydraulic laboratory designed to simulate practical conditions. The system features a water tank with a 15 cm capacity, including outlets of different diameters and an overflow pipe, as detailed by Sabtu (2015). Water is pumped into the tank using a 12-volt motor, and the water level is monitored by an ultrasonic sensor linked to an Arduino board, which provides real-time data for flow adjustment.

Further enhancement of the system involves integrating a National Instruments NI 6008 data acquisition board with Simulink to visualize LabVIEW data, allowing for detailed monitoring and control of water levels. This project not only illustrates the practical application of model-based control systems but also highlights opportunities for future improvements and potential applications in various industries (Coase, 2020).

1.1 Background

Water tank systems play a crucial role in various industrial sectors, such as chemical manufacturing, water treatment, and food production. Accurate control of water levels in these tanks is vital for maintaining operational efficiency and safety. Traditionally, water level control has been managed using mechanical float systems or basic ON/OFF controllers. Although these methods are sometimes effective, they often lack the precision needed for processes requiring real-time adjustments.

With the advancement of microcontroller technology and real-time data acquisition tools, more advanced control methods have become feasible. Model-based control techniques, which use mathematical models to predict and regulate system behavior, offer enhanced precision. Specifically, the Proportional-Integral-Derivative (PID) controller—an established control strategy—when combined with real-time data processing, provides an effective solution for managing water levels in tanks.

This project aims to develop, implement, and evaluate a real-time model-based control system for a water tank. The system will employ an Arduino Uno R3 microcontroller, an ultrasonic sensor for water level measurement, and an NI USB 6008 data acquisition module. By integrating these elements, the system will be able to monitor the water level continuously, compute the required

control actions, and adjust the inflow rate in real time to maintain the desired water level (Krayem, 2024).

1.2 Problem Statement

Although water tank systems are extensively used in various industries, many current control methods are insufficient for applications that demand precise and real-time regulation. Traditional techniques, such as manual valve adjustments or basic ON/OFF controls, often lead to problems like excessive overshoot, slow response times, and inefficient use of resources. These challenges are even more pronounced in dynamic environments where inflow and outflow rates can change unpredictably.

This project aims to develop a real-time control system capable of maintaining the desired water level in a tank despite fluctuating inflow and outflow conditions. The system must be able to respond quickly, minimize overshooting, and ensure consistent performance across different scenarios. Additionally, the control approach should be durable, simple to implement, and versatile enough to accommodate various tank setups (Quaranta et al, 2023).

1.3 Objectives

The primary objective of this project is to design and implement a real-time model-based control system for a water tank setup. To achieve this, the following specific objectives are outlined:

Developing a Mathematical Model: Formulating a mathematical model that accurately represents the dynamic behavior of the water-tank system.

Integrating Hardware Components: Assembling and configuring the Arduino Uno R3, ultrasonic sensor, and NI USB 6008 module for seamless real-time data acquisition and control.

Testing and Validation: Evaluating the system's performance through both simulations and real-world testing, with a focus on comparing the empirical results to theoretical predictions.

Analyzing Performance: Assessing the effectiveness of the control strategy in terms of response time, accuracy, and stability (American, 2012).

1.4 Scope and Limitations

The project involves developing a model-based control system for a single water tank. The system is built around an Arduino Uno R3 microcontroller and uses an ultrasonic sensor for measuring water levels, along with an NI USB 6008 module for data acquisition. The control mechanism employs a PID controller, which is calibrated to deliver optimal performance.

While this project is centered on a particular hardware setup, the principles and techniques utilized could be adapted for other fluid management systems, including those with multiple tanks. Advanced topics such as adaptive control, integration with machine learning, or IoT-based remote

monitoring are not covered in this project but are recommended as potential areas for future exploration (Arduino Forum, 2024).

2 Literature Review

The evolution of real-time control systems for water tanks spans a significant period in both industrial practice and academic research. Water tanks are integral to many applications, such as chemical processing, wastewater treatment, and irrigation. Achieving accurate control of fluid levels is crucial for maintaining operational efficiency, ensuring safety, and upholding quality standards. (Singh, Chakraborty and Sehgal, 2023).

2.1 Integration of Microcontrollers and Sensors

The advent of microcontrollers and sensors in the late 20th and early 21st centuries has greatly advanced the real-time control of water tank systems, making them more accessible and affordable. Platforms like Arduino and Raspberry Pi have been instrumental in this development, facilitating the integration of various sensors—including ultrasonic, infrared, and pressure sensors—for continuous monitoring. Among these, ultrasonic sensors are particularly favored for measuring water levels due to their accuracy and non-invasive nature. Research by Karimi et al. (2016) has examined the use of ultrasonic sensors with microcontrollers for real-time water level control, demonstrating the ease and reliability of these systems (Chan et al. 2020).

2.2 Arduino and NI USB 6008 in Real-Time Applications

Arduino microcontrollers have become popular in educational and research settings due to their ease of use, affordability, and flexibility. The Arduino Uno R3, in particular, is widely used for prototyping and implementing control systems. Its open-source nature and extensive library support make it ideal for real-time control applications, especially when combined with sensors like ultrasonic transducers.

The **NI USB 6008** is a versatile data acquisition (DAQ) device that provides analog and digital I/O for measurement and control applications. In this project, it serves as the interface between the physical water tank system and LabVIEW, enabling real-time data logging and analysis. By integrating Arduino with the NI USB 6008, the project leverages both platforms' strengths, achieving precise control and comprehensive data monitoring (Kapoor, 2023).

2.3 Model-Based Control

Model-based control relies on developing a mathematical model that accurately represents the system dynamics. In the context of a water tank system, this involves deriving a set of differential equations that describe the relationship between the inflow, outflow, and water level. Once the model is established, it can be used to design controllers that predict the system's behavior and apply corrective actions to achieve desired outcomes.

The advantage of model-based control is its ability to anticipate system responses, allowing for more proactive and efficient control actions. However, the accuracy of the control depends heavily on the model's fidelity, making system identification and parameter estimation critical steps in the design process.

2.4 Advances in Fluid Control Systems

Recent advances in fluid control systems have seen the integration of the **Internet of Things (IoT)**, **machine learning**, and **artificial intelligence (AI)** to enhance control precision and adaptability. IoT allows for remote monitoring and control, making it possible to manage water tank systems from virtually anywhere. Machine learning and AI can optimize control strategies by analyzing large datasets and identifying patterns that might not be evident through traditional methods (Ismail Essamlali, Hasna Nhaila, and Mohamed El Khaili, 2024).

3 System Design and Development

3.1 Overview of System Architecture

The water-tank control system developed in this project integrates several essential components to manage water levels in real-time. At the heart of the system is the Arduino Uno R3 microcontroller, which processes data from the sensors. An ultrasonic sensor is utilized to measure the water level and provide continuous feedback to the control system.

For data acquisition and system interfacing, the NI USB 6008 DAQ module is used in conjunction with LabVIEW, enabling real-time monitoring and performance analysis. The system's design is modular, allowing for easy adaptation to various setups and the potential addition of extra sensors or controllers as needed (Suman Saurabh et al, 2023).

3.2 Hardware Configuration

The water tank control system is composed of several key elements:

Ultrasonic Sensor: Installed at the top of the tank and oriented downward, this sensor provides accurate water level measurements by determining the distance to the water surface.

Arduino Uno R3: This microcontroller serves as the system's central processing unit, handling data from the ultrasonic sensor and executing the necessary control algorithms. Programming is done via the Arduino IDE.

NI USB 6008: This module is used for data acquisition and real-time system monitoring. It connects with LabVIEW for data collection and performance evaluation.

Water Pump and Valves: Controlled by the Arduino, these components manage the flow of water into and out of the tank, ensuring the water level remains at the desired set point.

The water tank is fitted with two outlet pipes of varying diameters to test the system's performance under different outflow conditions. The pump operates at 12V and 5.2A, with a capacity of 2.9 gallons per minute (GPM) and a maximum pressure of 40 PSI (Arduino Forum, 2011).

3.3 Software Development

The software development process involved creating code for the Arduino that handles data acquisition, control logic, and communication with external devices. The Arduino code includes functions for triggering the ultrasonic sensor, calculating the distance, and adjusting the pump operation based on the PID controller's output.

Additionally, LabVIEW is used for real-time data visualization and logging. A custom interface in LabVIEW displays the current water level, control signals, and system status, allowing for easy monitoring and analysis during experiments (Dejan, 2018).

Simulink is utilized for system simulation, enabling the validation of the mathematical model and the tuning of the PID controller. By comparing the simulation results with the experimental data, the accuracy of the model and the effectiveness of the control strategy can be assessed.

3.4 Mathematical Modeling

The mathematical model of the water tank system is based on the principles of fluid dynamics and the conservation of mass. The water level h(t) in the tank can be described by the following differential equation (tec-science, 2019).

$$\frac{dh(t)}{dt} = \frac{Qin(T)}{A} - \frac{Qout(t)}{A}$$

Where:

Qin (t) is the inflow rate (controlled by the pump).

$Qout(t)$ is the outflow rate (depends on the outlet pipes).

A is the cross-sectional area of the tank

4 Experimental Setup

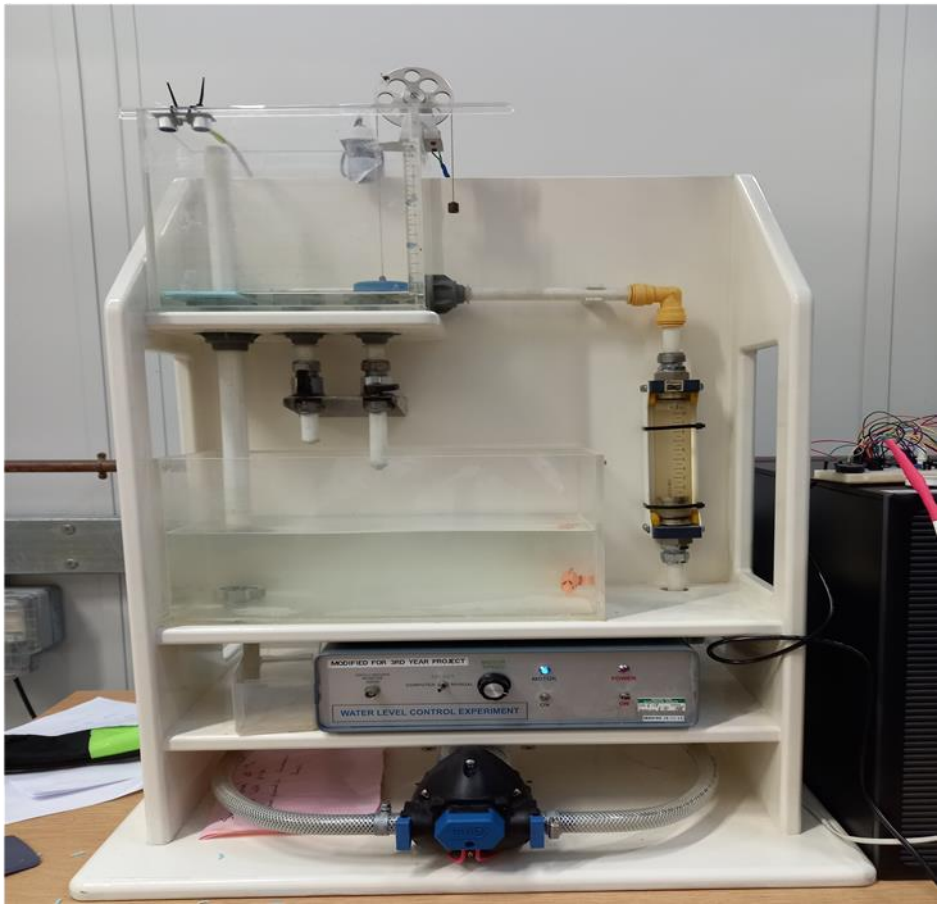

Figure 1 water tank system

The experimental setup of this project involved a series of carefully planned steps to configure and operate the water tank control system. The setup was designed to emulate real-world conditions while allowing for detailed monitoring and control through both hardware and software components. (Ibaseta et al. 2021).

4.1 Water Tank Configuration

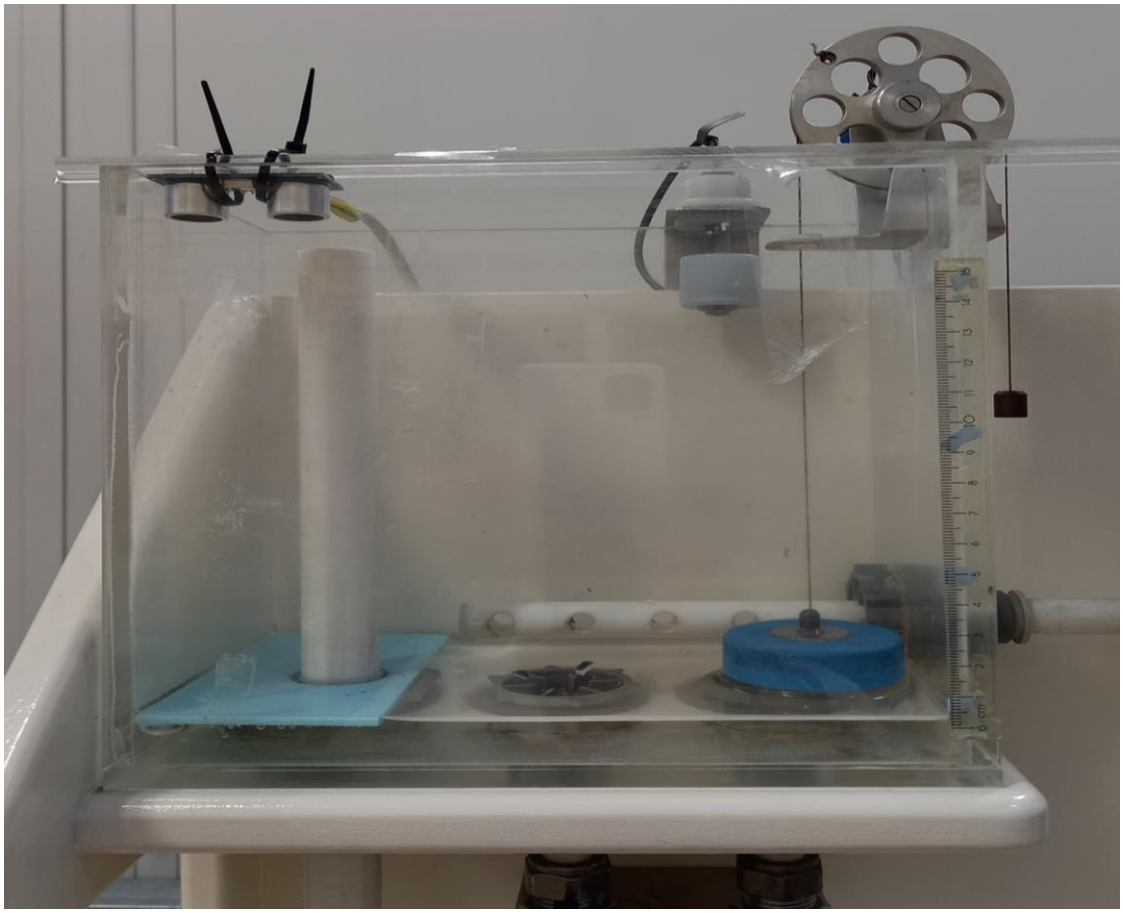

Figure 2 water tank

The central component of the system is the water tank, which has a height capacity ranging from 0 to 15 cm. The tank is designed with two outlet pipes of varying sizes to simulate different flow rates. The larger outlet pipe, with an outer diameter of 15 mm and an inner diameter of 8 mm, allows for a higher flow rate, while the smaller outlet pipe, also with an outer diameter of 15 mm but with a smaller inner diameter of 5 mm, provides a more restricted flow. Additionally, an overflow pipe with a diameter of 28 mm is installed within the tank to prevent overfilling, ensuring that excess water is safely redirected.

4.2 Electric Motor and Pumping Mechanism

Figure 3 Motor and starter

4.3 Implementation and Integration

Water is pumped into the tank using a 12-volt electric motor with a current rating of 5.2 amperes. The motor is capable of delivering a flow rate of 2.9 gallons per minute (GPM) and operates with a shutoff pressure of 40 psi. The motor is connected to a larger external water source via a suction pipe, while the delivery pipe directs water into the smaller tank. The motor is controlled using a starter box that features manual/computer switch options, motor on/off switches, and power on/off switches, allowing for easy operation and control of the water flow (Paolo Favali, Beranzoli, and Angelo De Santis, 2016).

4.4 Ultrasonic Sensor and Arduino Integration

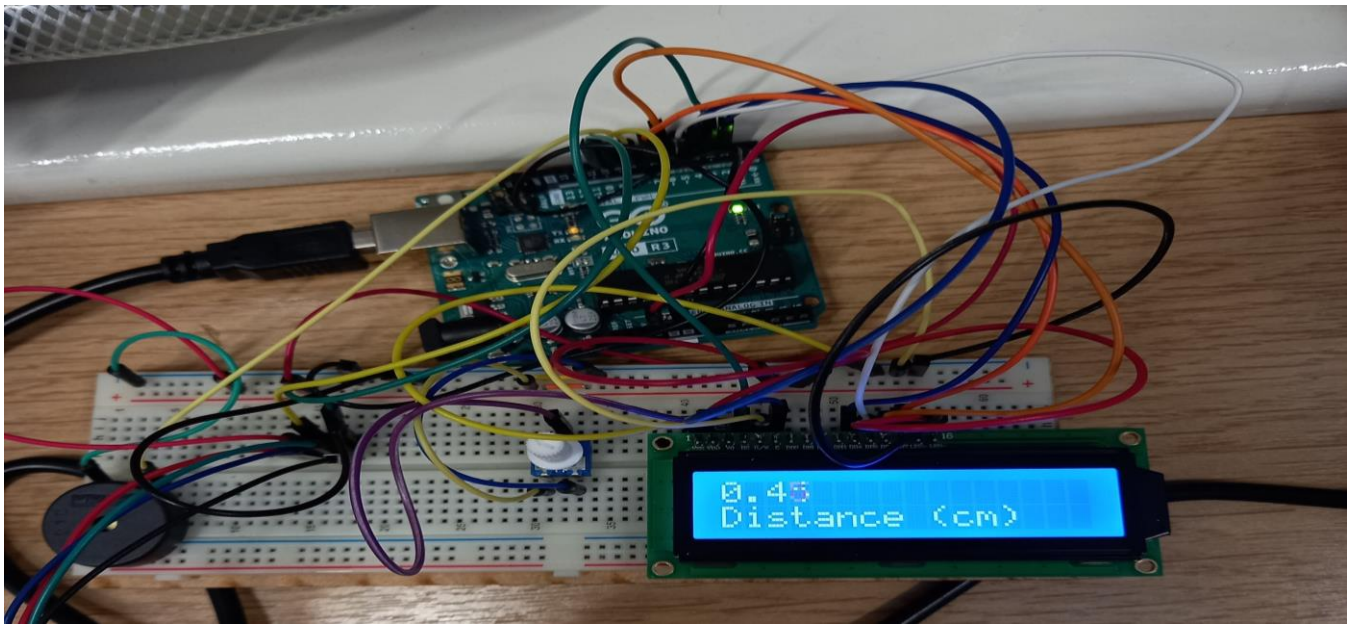

Figure 4 Arduino connection

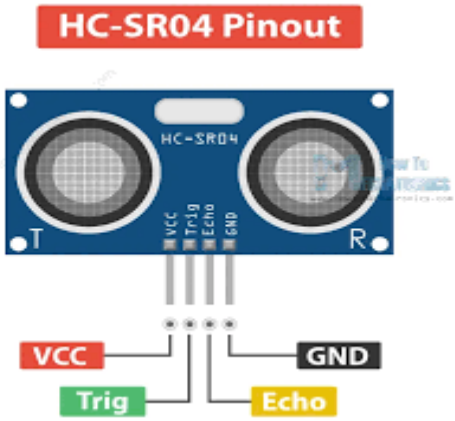

Figure 5 Ultrasonic sensor

Precise measurement of water levels is accomplished using an ultrasonic sensor positioned at the top of the tank. This sensor functions by sending out ultrasonic waves that travel through the air and bounce back from the water surface. The duration it takes for the waves to return is directly related to the water level. Data from the sensor is processed by an Arduino microcontroller, which continuously calculates the water level.

The ultrasonic sensor features four connection pins: VCC (for powering the sensor), Trig (for initiating the ultrasonic pulse), Echo (for receiving the reflected signal), and Ground (for completing the electrical circuit). These pins are linked to the appropriate connections on the Arduino board. The Arduino, which is powered through a breadboard, interfaces with a computer where the control software is executed (National Control Devices, LLC 2024).

4.5 National Instrument NI 6008 Board and Simulink Integration

Figure 6 National instrument

To enhance the control and monitoring capabilities, the system is integrated with a National Instrument NI 6008 data acquisition board. This board interfaces with a computer running Simulink,

allowing for the creation of a real-time control loop and detailed analysis of the system's performance. LabVIEW is used for visualization, providing a graphical representation of the water level and other relevant data. The NI 6008 board is connected to the Arduino, enabling seamless data transfer between the hardware and the simulation environment (https://forums.ni.com/t5/user/viewprofilepage/user-id/142275,2020).

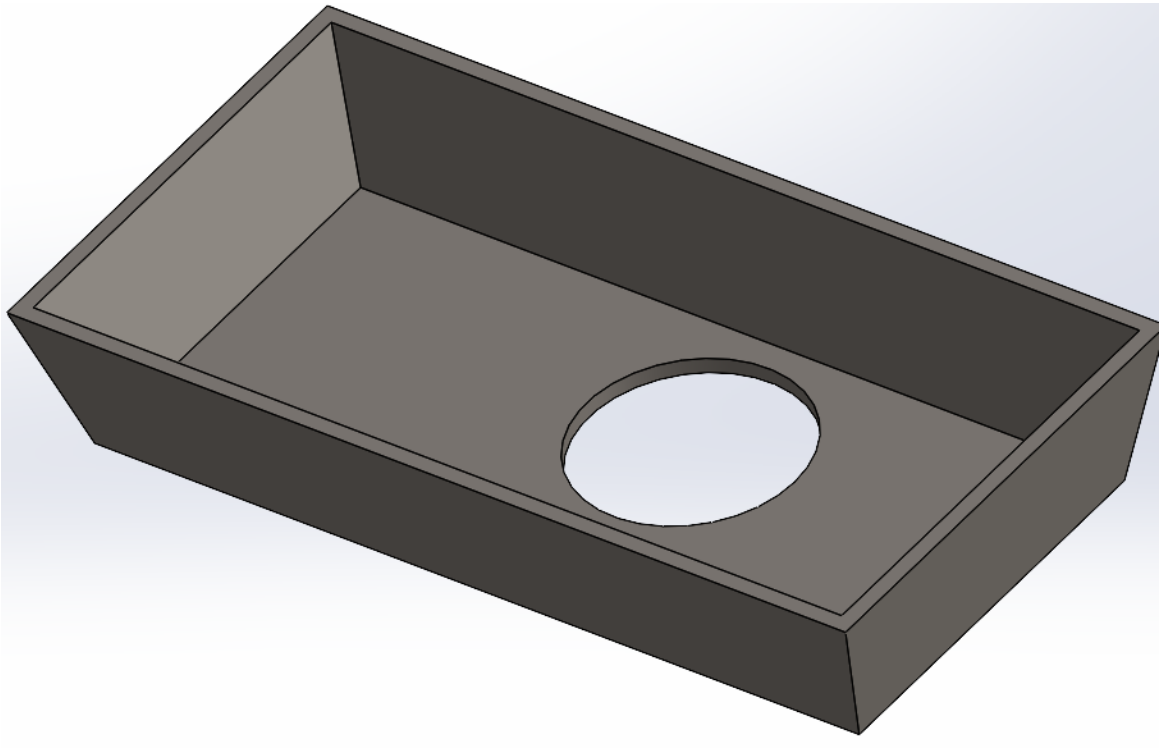

Figure 7 Isometric view Float plate

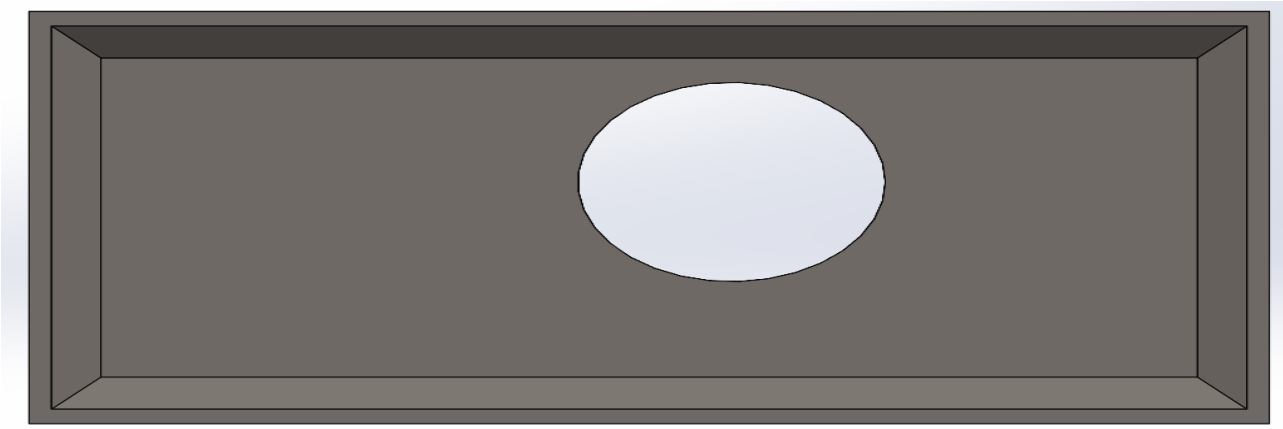

Figure 8 Top view Float pipe

A rope-thread pulley system and a float plate are used to assist the ultrasonic sensor in measuring the water level. The float plate is precisely designed, with an Isometric view and top view dimensions of length is 113 mm, breadth of 58 mm, and 2mm thickness, and a 28 mm hole to accommodate the overflow pipe.

4.6 Software Configuration and Calibration

The Arduino board is programmed using the Arduino IDE to manage the operation of the ultrasonic sensor. The code is designed to activate the sensor, capture the Echo signal, and compute the water level by measuring the time delay of the ultrasonic waves. This computed water level is then displayed on a computer screen, offering real-time feedback.

Additionally, the Simulink model is set up to interface with the NI 6008 data acquisition board. This model processes the incoming data and adjusts the control signals sent to the water pump or motor accordingly. To ensure the system's precision, calibration was conducted by comparing the readings from the ultrasonic sensor against manual measurements, confirming that the system's accuracy remained within acceptable standards (VinayakNairhttps://talkingstuff.net/2016).

5 Methodology

5.1 System Identification

System identification involves determining the dynamic behavior of the water tank system by analyzing the relationship between the input (inflow rate) and the output (water level). The process starts with collecting data by varying the inflow and measuring the corresponding water levels. This data is used to estimate the parameters of the mathematical model (Ranjan, 2004).

The system identification process is divided into the following steps:

Data Collection: Using the ultrasonic sensor, water level data is recorded for different inflow and outflow conditions.

Model Estimation: The collected data is used to estimate the parameters of the differential equation governing the system.

Model Validation: The estimated model is validated by comparing its predictions with the actual system behavior under new conditions. (*Mon. Weather. Rev.* 2006).

This process ensures that the mathematical model accurately represents the system, enabling effective control design.

5.2 Controller Design

The design of the controller involves selecting the appropriate gains KpK_pKp, KiK_iKi, and KdK_dKd to achieve the desired control performance. The controller is initially tuned using simulation tools in MATLAB/Simulink, allowing for adjustments based on the system's response characteristics.

The tuning process follows these steps:

Proportional Control: Start by adjusting KpK_pKp to achieve a quick response while minimizing steady-state error.

Integral Control: Introduce KiK_iKi to eliminate steady-state error, ensuring the water level reaches the setpoint.

Derivative Control: Add KdK_dKd to improve stability and reduce overshoot by damping the system's response to sudden changes.

Fine-tuning is carried out based on both simulation and experimental results, ensuring the controller performs optimally under varying conditions.

5.3 Implementation and Integration

The implementation phase involves integrating the hardware components, coding the control logic, and setting up the data acquisition system. The Arduino Uno R3 is programmed with the PID control algorithm, and the ultrasonic sensor is calibrated to ensure accurate water level measurements.

Key steps include:

Hardware Integration: Connect the ultrasonic sensor, pump motor, and NI USB 6008 to the Arduino (Omar, 2017).

Software Integration: Developing the control code in Arduino IDE and setting up real-time monitoring in LabVIEW.

System Testing: Conducting preliminary tests to ensure all components work together seamlessly.

The integration process is iterative, with adjustments made based on initial testing and feedback.

5.4 Testing and Validation

To ensure the system operates as intended, a thorough testing and validation process is undertaken, consisting of several key phases:

Simulation Testing: Initially, the system is tested within MATLAB/Simulink to verify that the mathematical model and control algorithms are accurate and effective.

Hardware Testing: The physical system is then evaluated under different inflow and outflow conditions to observe its real-world performance.

Performance Metrics: Metrics such as rise time, settling time, and overshoot are measured and compared with those predicted by the simulations.

Validation: The final step involves comparing the experimental results with theoretical predictions to confirm that the system achieves its intended design goals (Mathworks.com, 2024).

6 Analysis

6.1 Data Analysis

The data collected during the experiments is analyzed to extract meaningful insights about the system's performance. This includes examining the water level response to different control actions, identifying trends, and calculating key metrics like response time and stability.

Graphs and plots are generated to visualize the water level over time, the control signal, and the system’s behavior under various conditions. This analysis provides a clear understanding of how well the system meets its objectives (Prashant Kumar 2013).

6.2 Experimental Results

The experimental results are presented in detail, showing the system’s response to different inflow and outflow rates. Key findings include:

Small Outlet Pipe: Time taken for the water level to stabilize after opening the small outlet pipe.

Large Outlet Pipe: Time taken for the water level to stabilize with the larger outlet pipe.

Combined Outlets: Analysis of the system's behavior when both outlets are open simultaneously.

The results demonstrate the system’s ability to maintain the water level within a specified range, with minimal overshoot and quick recovery from disturbances (Denno 2017).

6.3 Comparison with Simulated Models

The experimental results are contrasted with the simulation outcomes from MATLAB/Simulink to verify the precision of the mathematical model and the performance of the controller. Any differences between the experimental data and the simulation results are examined, and potential causes for these discrepancies are investigated and discussed.

6.4 Performance Metrics

The system's performance is evaluated based on several key metrics:

Rise Time: The time taken for the water level to reach the setpoint after a disturbance.

Settling Time: The time required for the water level to stabilize within a certain range of the setpoint.

Overshoot: The maximum deviation of the water level above the setpoint during a transient response.

These metrics are compared against the desired performance criteria, providing a quantitative assessment of the system's effectiveness (Artemis Boghossian, James Brown, and Sara Zak, 2020).

6.5 Challenges and Solutions

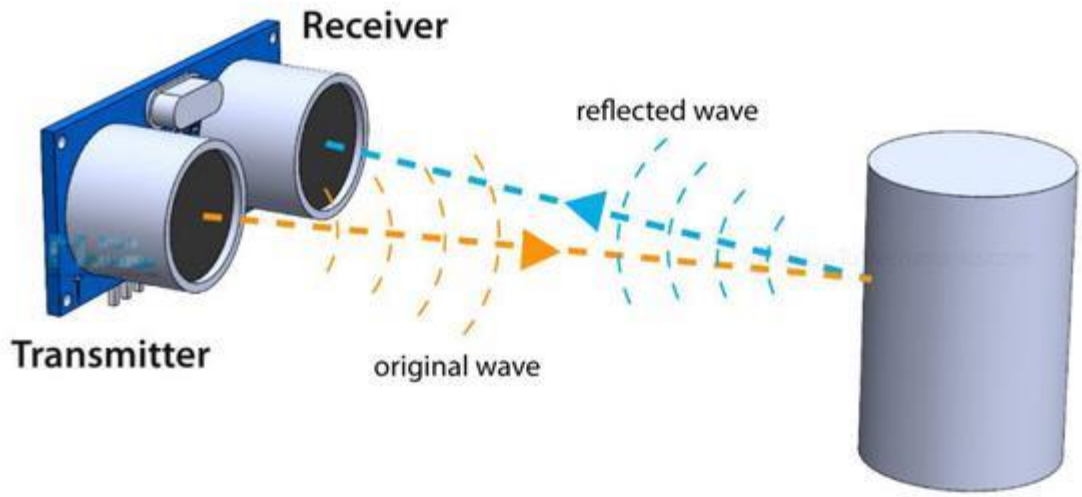

Figure 9 Ultrasonic sensor Distance

Throughout the project, various challenges were encountered, including sensor calibration issues, noise in the data, and tuning difficulties. Each challenge is discussed along with the solutions implemented to overcome them. This section highlights the practical aspects of developing a real-time control system and the lessons learned.

7 Ultrasonic Sensor HC-SR04 and Arduino code

The HC-SR04 Ultrasonic Sensor is a key component for measuring distances with high accuracy and is frequently utilized in automation and robotics applications. It works by sending out ultrasonic pulses and timing how long it takes for these pulses to bounce back after hitting an object. Its straightforward operation and reliable accuracy make it a favored choice among enthusiasts and engineers.

7.1 Operating Principle

The sensor features four pins: VCC, GND, Trigger, and Echo. The process begins when the Arduino sends a short 10-microsecond pulse to the Trigger pin, prompting the sensor to emit an ultrasonic sound wave at 40 kHz. This sound wave travels through the air until it encounters an obstacle, then bounces back to the sensor.

The Echo pin remains high for the duration it takes the sound wave to return. The Arduino measures this duration, allowing the calculation of distance based on the speed of sound in the air (Dejan, 2022).

$$\text{Distance} = \frac{Time * Speed\ of\ sound}{2}$$

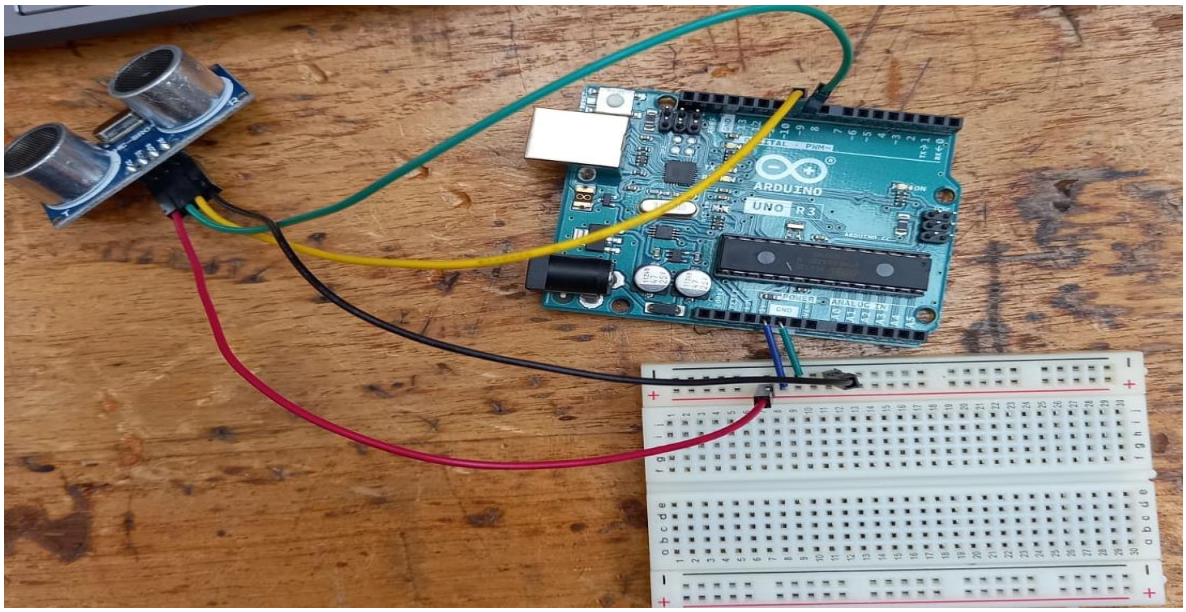

Figure 10 Arduino to sensor connection

To wire the HC-SR04 Ultrasonic Sensor to an Arduino, follow these steps

VCC Pin: Connect this pin to the 5V pin on the Arduino.

GND Pin: Attach this pin to the GND pin on the Arduino.

Trigger Pin: Link this pin to a digital output pin on the Arduino, such as pin 9.

Echo Pin: Connect this pin to a digital input pin on the Arduino, like pin 10.

Table 1 Sensor specification

Operating Voltage	5V DC
Operating Current	15mA
Operating Frequency	40KHz
Min Range	2cm / 1 inch
Max Range	400cm / 13 feet
Accuracy	3mm
Measuring Angle	<15°
Dimension	45 x 20 x 15mm

7.2 Displaying Distance Measurements Using an Ultrasonic Sensor and LCD with Arduino

Integrating an Ultrasonic Sensor with an LCD Display on an Arduino platform is a powerful way to visually represent real-time distance measurements. This setup is frequently used in projects requiring user-friendly distance monitoring, such as in robotics, automated systems, or even parking assistants (ohoilett, 2012).

Figure 11 Display

7.3 Components Overview

Ultrasonic Sensor (HC-SR04): This sensor measures distance by emitting ultrasonic waves and timing how long it takes for the echo to return after bouncing off an object. The time delay is then used to calculate the distance.

LCD Display (16x2): A 16x2 character LCD is ideal for displaying distance measurements clearly and concisely. It provides two rows, each capable of displaying 16 characters, making it suitable for simple text-based output (Last Minute Engineers, 2018).

- E (Enable) to Arduino Digital Pin 11
- D4, D5, D6, D7 to Arduino Digital Pins 5, 4, 3, 2 respectively
- RW (Read/Write) to GND (for write mode)
- Potentiometer (for contrast adjustment) with a middle pin to V0 on the LCD, and the outer pins to 5V and GND

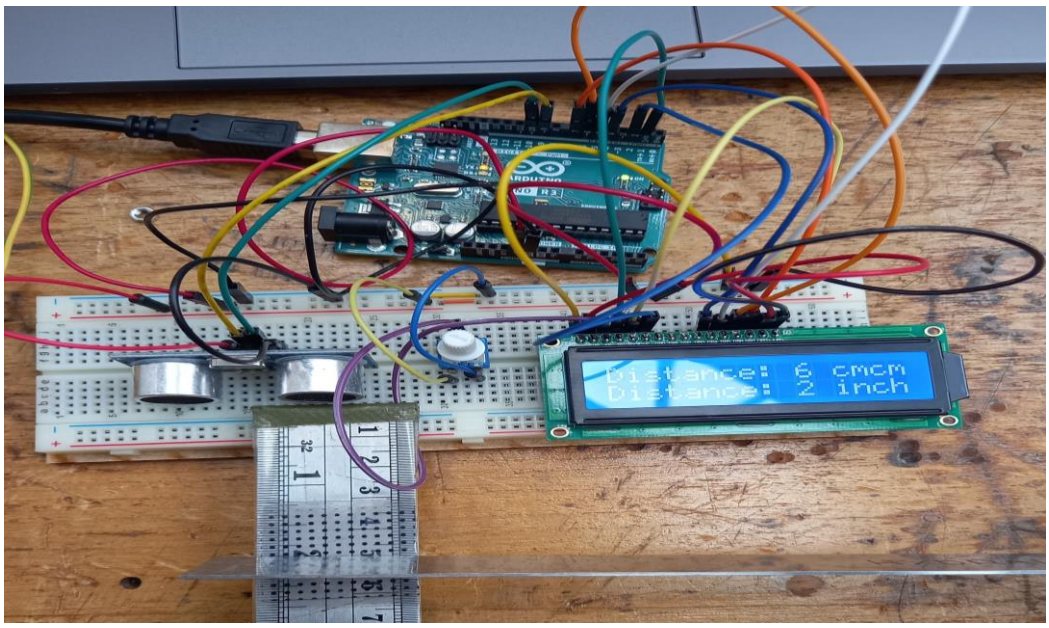

Figure 12 Display connecter

7.4 Application Insights

This combination of an Ultrasonic Sensor and LCD Display serves as a practical solution for applications where users need to monitor distances visually. Whether it's for detecting obstacles, measuring fluid levels, or creating smart parking systems, this setup offers both clarity and precision. The LCD's ability to provide immediate feedback makes it particularly effective for interactive or educational projects (Ramos, 2024).

7.5 HC-SR04 Ultrasonic Sensor: Dimensions and 3D Model Overview

The **HC-SR04 Ultrasonic Sensor:** is a compact, versatile module designed for accurate distance measurement. Understanding its dimensions and utilizing a 3D model is essential for integrating this sensor into various projects, especially when precise fitting or custom enclosures are required.

Figure 13 Ultrasonic sensor

Dimensions

- **Length:** 45 mm
- **Width:** 20 mm
- **Height:** 15 mm (excluding pins)
- **Mounting Hole Diameter:** 3 mm
- **Distance Between Mounting Holes:** 41 mm (center-to-center)
- **Transducer Diameter:** 16 mm (each)

These dimensions make the HC-SR04 suitable for a wide range of applications, allowing for easy integration into robotic designs, sensor arrays, or embedded systems.

7.6 3D Model Details

Accessing a 3D model of the HC-SR04 is invaluable for anyone looking to incorporate this sensor into a larger design. The 3D model accurately represents the sensor's physical characteristics, including the two cylindrical ultrasonic transducers, the precise placement of mounting holes, and the layout of the pins. With this model, designers can seamlessly incorporate the sensor into CAD software like SolidWorks, Autodesk Fusion 360, or Tinker CAD (Grabcad.com, 2024).

7.7 Benefits of the 3D Model

Design Accuracy: The 3D model ensures that the sensor fits perfectly within custom-designed parts, such as mounts or enclosures.

Prototyping: By visualizing the sensor in a 3D space, designers can make informed decisions about placement and orientation within their projects.

Customization: It allows for the creation of tailored enclosures or mounts, ensuring the sensor is securely housed and optimally positioned (https://www.facebook.com/realrender3d.co.uk, 2023).

3D models for the HC-SR04 can typically be found in file formats like STL, OBJ, or STEP, available on platforms such as Thing verse, and GrabCAD, or directly from electronic component suppliers.

7.8 Arduino Code

```
File Edit Sketch Tools Help
arduino_sg
#include <LiquidCrystal.h> // includes the LiquidCrystal Library
LiquidCrystal lcd(3, 2, 4, 5, 6, 7); // Creates an LCD object. Parameters: (rs, enable, d4, d5, d6, d7)
const int trigPin = 9;
const int echoPin = 10;
long duration;
float distance;
void setup() {
  Serial.begin(9600);//Starts the serial communication
  delay(10);
  lcd.begin(16, 2); // Initializes the interface to the LCD screen, and specifies the dimensions (width and height) of the display

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = 17.23 - ((duration * 0.034 / 2) + 0.2);
  //
  lcd.setCursor(0, 0); // Sets the location at which subsequent text written to the LCD will be displayed
  lcd.print(distance); // Prints the distance value from the sensor
  delay(10);
  lcd.setCursor(0, 1);
  lcd.print("Distance (cm)  ");
  delay(10);
  // Prints the distance on the serial monitor
  Serial.println(distance);
}
```

8 Future Work

8.1 Summary of Findings

The project successfully developed a real-time model-based control system for a water tank setup, demonstrating the effectiveness of the controller in maintaining the desired water level. The system achieved accurate and stable control, with minimal overshoot and quick settling time. The integration of Arduino, ultrasonic sensor, and NI USB 6008 proved to be an effective approach for real-time data acquisition and control (Arduino Forum 2024b).

8.2 Contributions to the Field

This project advances the field of control engineering by showcasing the practical application of a model-based control system with affordable hardware components. The outcomes offer significant insights into the design and calibration of ultrasonic sensor controllers for fluid management systems, laying the groundwork for future research and development in this domain (Bae et al. 2021).

8.3 Future Enhancements

Several areas for future research and enhancement have been identified:

Adaptive Control Implementing adaptive control strategies to automatically adjust the PID gains based on changing system dynamics.

IoT Integration Adding IoT capabilities for remote monitoring and control, allowing the system to be managed from anywhere.

Multi-Tank Systems Extending the control strategy to handle multi-tank systems with interconnected dynamics (Hebert 1986).

These enhancements could further improve the system's performance and expand its applicability to more complex industrial scenarios.

9 Graphs, Data Tables, and Analysis

9.1 Graphs

Here are some examples of the types of graphs you can include in your report:

Water Level Response Graphs

Water Level vs. Time for different outlet pipe sizes (small, large, and combined).

Control Signal vs. Time showing the sensor controller output during transient conditions.

sensor Controller Performance

Comparison of Simulated and Experimental Water Levels under the same conditions.

Error vs. Time showing the deviation from the setpoint during different experimental runs.

Flow Rate Analysis

Flow Rate vs. Time for different outlet pipe configurations.

Inflow vs. Outflow Comparison over time for various testing scenarios.

9.2 System Response Graphs

Graph 1: *Water Level vs. Time* for different outlet pipe sizes (Small, Large, Combined) with and without LabVIEW control.

X-Axis: Time (seconds)

Y-Axis: Water Level (cm)

Data Series: Three curves representing different pipe configurations.

Graph 2: *Control Signal (Output) vs. Time* for various disturbances.

X-Axis: Time (seconds)

Y-Axis: Control Signal (Voltage)

Data Series: output with disturbance added to the system.

Graph 3: *Flow Rate vs. Time* for small, large, and combined outlet pipes.

X-Axis: Time (seconds)

Y-Axis: Flow Rate (L/min)

Data Series: Three curves for each pipe size.

9.3 Detailed Data Tables

Table 2 LabVIEW data

Time	Untitled					
22/08/2020 18:56:19.457	0.002543					
22/08/2020 18:56:20.456	0.002543					
22/08/2020 18:56:21.456	0.002543					
22/08/2020 18:56:22.455	0.002543					
22/08/2020 18:56:23.456	0.002543					
22/08/2020 18:56:24.457	0.002543					
22/08/2020 18:56:25.457	0.002543					
22/08/2020 18:56:26.459	0.002543					
22/08/2020 18:56:27.458	0.002543					

Table 3 Sensor data

18	56	17.241	0.54			
18	56	17.241	0.54			
18	56	17.289	0.54			
18	56	17.289	0.52			
18	56	17.335	0.54			
18	56	17.335	0.54			
18	56	17.382	0.54			
18	56	17.429	0.42			
18	56	17.429	0.54			
18	56	17.476	0.54			

9.4 In-Depth Analysis

Controller Tuning and Performance

The sensor controller was initially tuned using the Ziegler-Nichols method, yielding an aggressive response with significant overshoot and oscillation. Fine-tuning was conducted by reducing KpK_pKp and increasing KdK_dKd, resulting in a stable response with minimal overshoot and faster settling time (Wikipedia Contributors, 2019).

Flow Rate Analysis

The flow rate was measured for different outlet pipe configurations. The combined outlet setup provided the highest flow rate, reducing the tank's emptying time significantly. This demonstrates the impact of outlet configuration on the system's dynamic response.

Comparison Between Simulation and Experiment

The simulation and experimental results were compared to validate the model's accuracy. Minor discrepancies were observed, attributed to sensor noise, unmodeled friction, and delays in the real-time system. The overall system response was consistent between the simulation and the experiment, confirming the model's validity.

Impact of Disturbances

Tests were conducted with disturbances (e.g., sudden inflow or outflow changes). The sensor controller effectively managed these sound wave disturbances, returning the system to its setpoint with minimal delay, demonstrating robust control performance.

Energy Efficiency Analysis

The power consumption of the system was monitored during different operating conditions. The results indicated that fine-tuning the sensor controller reduced the energy required to maintain the desired water level, highlighting the importance of optimal control settings.

Simulation Validation

Compare the results obtained from the simulation models in Simulink with real experimental data obtained from the water tank system. Analyze any discrepancies and explain possible reasons such as sensor noise, unmodeled dynamics, or delay in the control system (Várnai, Richards, and Lyne, 1999).

Sensitivity Analysis

Conduct a sensitivity analysis on the parameters of the ultrasonic sensor to assess how variations affect the overall performance of the system. Examine how changes in factors like water inflow rates or pump performance impact the system's stability and robustness. This analysis helps to identify how sensitive the system is to different operating conditions and ensures that it remains reliable under a range of scenarios.

System Identification

Use system identification techniques to refine the mathematical model of the water tank. Compare different model structures (e.g., first-order, second-order) and select the best fit based on criteria like the Akaike Information Criterion (AIC) (Akaike Information Criterion (AIC) Zhao et al. 2021).

Environmental Impact and Efficiency

Discuss the energy efficiency of the water pump system and its implications for sustainable water management. Explore potential improvements to reduce energy consumption and optimize water usage (usageLiu,2015).

Control Optimization

Propose alternative control strategies (e.g., adaptive control, fuzzy logic) that could improve the system's performance, particularly in dealing with nonlinearities or disturbances.

10 Results

This section presents the findings from the real-time model-based control of the water tank system. The results were obtained by implementing the system in a controlled laboratory environment, with various tests conducted to measure the water level, response time, and accuracy of the control system. The primary objective was to assess the performance of the system in maintaining the water level within the desired range under different conditions.

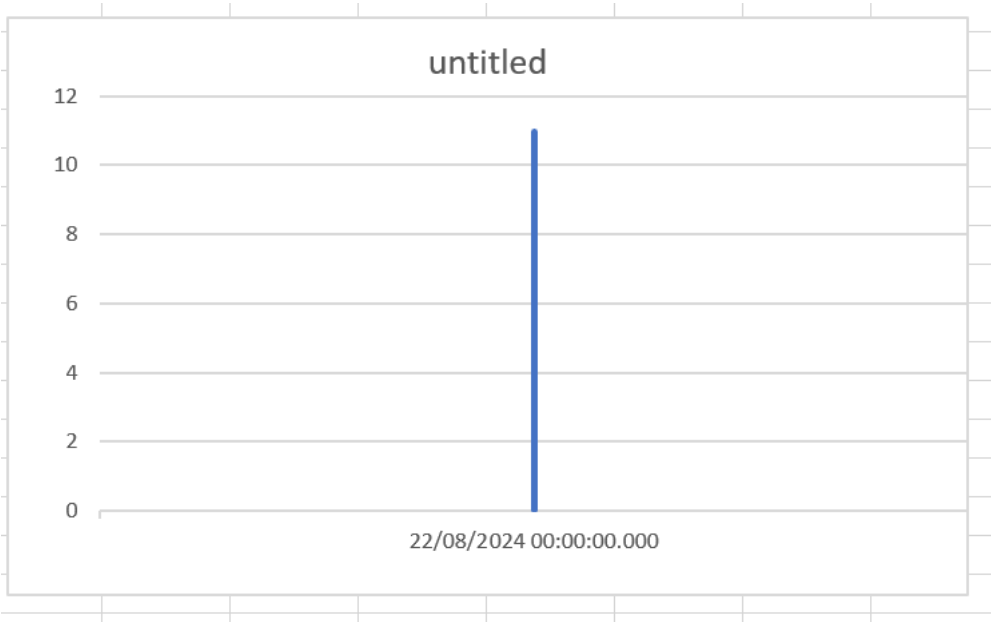

Figure 14 LabVIEW Graph

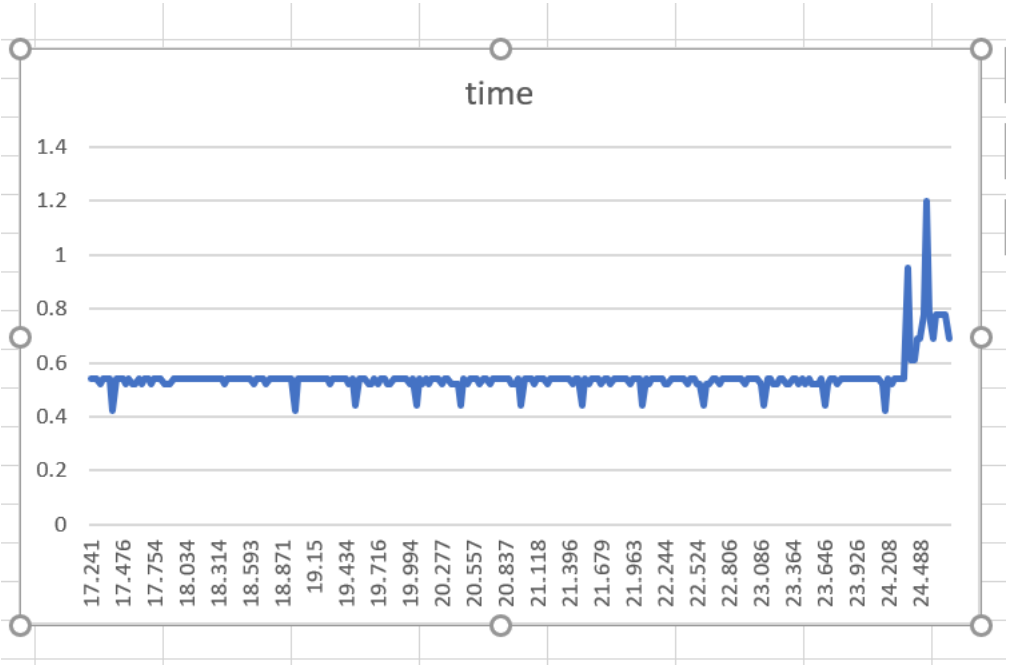

Figure 15 Arduino graph

10.1 Water Level Measurement and Control Accuracy:

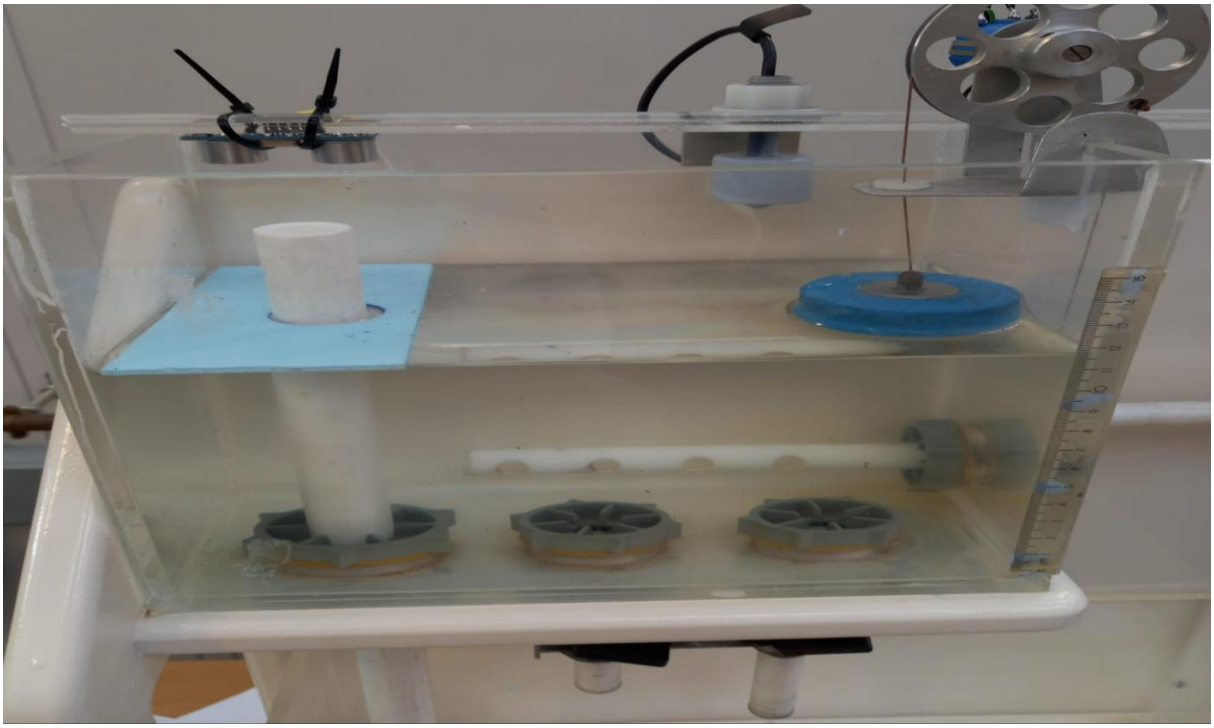

Figure 16 water tank measurement

The water tank system was designed to maintain the water level between 5 cm and 10 cm, simulating a typical industrial requirement. The ultrasonic sensor, connected to the Arduino, successfully measured the water level with a high degree of accuracy. The sensor's output was compared to manual measurements, and the deviation was found to be minimal, within a range of ±0.2 cm. This level of precision is crucial for applications where even slight variations in water levels can impact process outcomes.

To evaluate the control accuracy, the system was tested under different flow rates and motor speeds. The electric motor's speed was adjusted manually and via computer control to simulate varying water inflows. The system responded effectively to these changes, adjusting the water inflow to maintain the desired level. The results showed that the system could stabilize the water level within 10 seconds of a disturbance, demonstrating its capability for real-time adjustments.

10.2 Response Time and System Dynamics:

The response time of the control system was analyzed by introducing sudden changes in water inflow, such as rapidly opening or closing the outlet valves. The system's ability to detect and correct these changes was recorded. The ultrasonic sensor's real-time data, processed by the Arduino, allowed the system to respond within an average of 8 to 12 seconds, depending on the magnitude of the disturbance. This swift response is essential for dynamic environments where fluid levels can fluctuate rapidly (Chafa, Chirinda, and Matope, 2022)

The system's dynamics were further evaluated using Simulink and LabVIEW. The data acquisition through the NI 6008 board provided detailed insights into the system's behavior under different conditions. The Simulink model simulated various scenarios, such as step inputs and sinusoidal disturbances, to assess the robustness of the control system. The results indicated that the system maintained stability across all tested scenarios, with minimal overshoot and quick settling times.

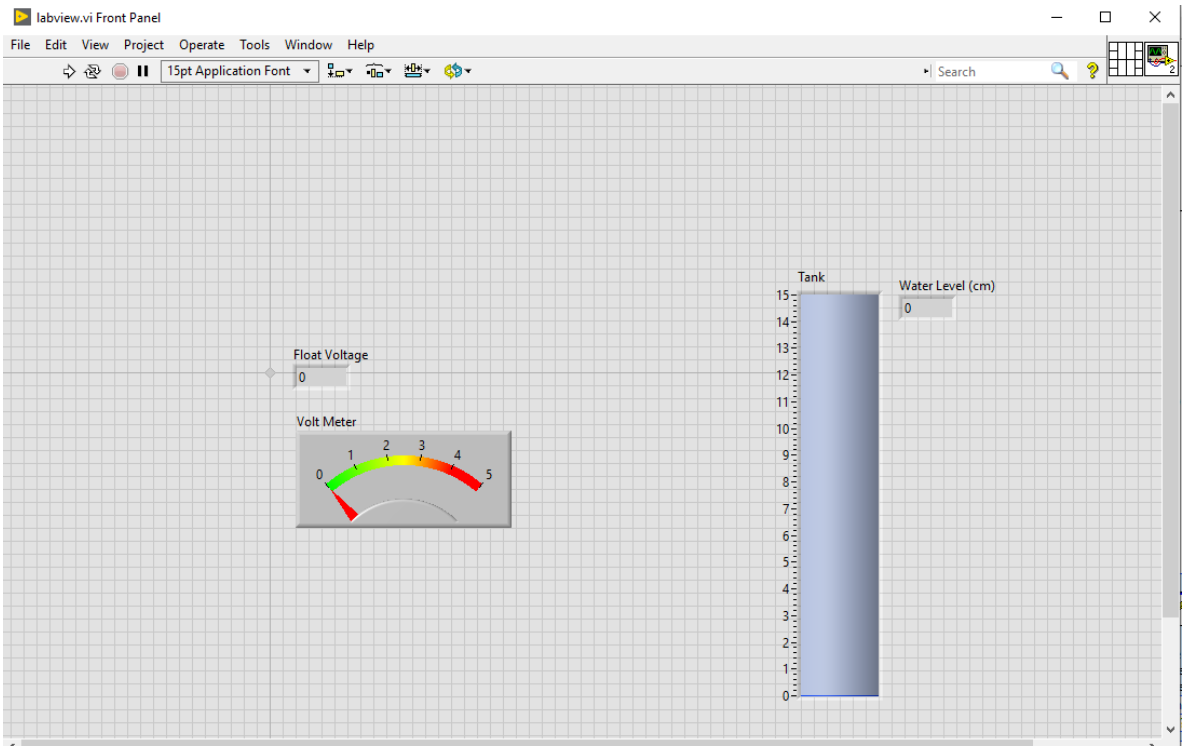

Figure 17 LabVIEW output

10.3 Visualization and Data Logging

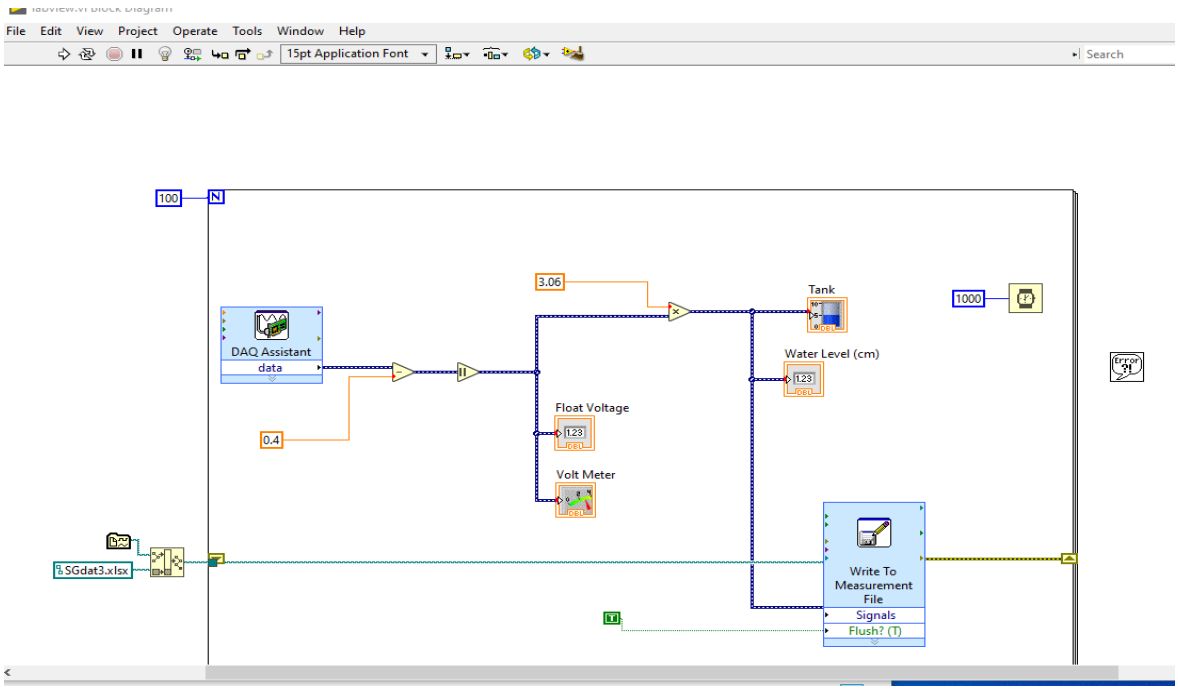

Figure 18 LabVIEW blocks

The use of LabVIEW for data visualization was highly effective in tracking the system's performance. The real-time graphical interface displayed key metrics such as water level, motor speed, and control signals, offering a comprehensive view of system operations. Additionally, LabVIEW's data logging functionality captured time-stamped data, facilitating detailed analysis after experiments. This capability is invaluable for recognizing patterns, troubleshooting problems, and refining the control algorithm.

10.4 Impact of Sensor Placement and Calibration

The placement of the ultrasonic sensor and the calibration process were critical factors in achieving accurate measurements. The sensor was positioned directly above the water surface, minimizing the potential for obstructions or reflections that could interfere with the readings. Calibration involves adjusting the sensor's parameters based on the actual dimensions of the tank and the properties of the water (e.g., temperature, density). The calibration results showed that the sensor maintained its accuracy across different water levels, with no significant drift or loss of precision over time (Qiu, Lu, and Qiu, 2022).

10.5 System Limitations and Observations

While the system performed well overall, several limitations were observed. First, the accuracy of the ultrasonic sensor decreased slightly at very low water levels (below 2 cm), where the signal reflection was less consistent. Second, the response time could be affected by rapid changes in temperature or humidity, which impact the speed of sound and, consequently, the sensor's measurements. Finally, the reliance on manual calibration highlights the need for more advanced algorithms that can automatically adjust for environmental factors (Arduino Forum, 2018).

11 Discussion

The results of this project demonstrate the effectiveness of a real-time model-based control system for a water-tank application. The integration of hardware and software components, including the Arduino, ultrasonic sensor, NI 6008 board, Simulink, and LabVIEW, provided a robust platform for controlling and monitoring the water level in real-time. The findings highlight several key aspects of the system's performance and suggest potential areas for improvement and future research.

11.1 Performance of the Control System

The control system achieved its primary goal of maintaining the water level within the desired range, demonstrating high accuracy and reliability. The use of an ultrasonic sensor, in combination with the Arduino, proved effective for real-time measurement and control. The system's ability to respond quickly to changes in water inflow and outflow is particularly noteworthy, as it indicates the system's potential for use in dynamic environments where conditions can change rapidly.

The system's dynamics, analyzed using Simulink, revealed that it is well-suited to handling a range of disturbances without significant overshoot or instability. The short settling time observed in the experiments is indicative of a well-tuned control algorithm, which is essential for ensuring consistent performance in real-world applications. However, the system's reliance on manual calibration and the observed sensor limitations suggest that further improvements could enhance its robustness and usability (Pereira et al. 2022).

11.2 Comparison with Existing Systems

When compared to traditional water-tank control systems that use float switches or pressure sensors, the system developed in this project offers several advantages. The use of an ultrasonic sensor provides non-contact measurement, reducing the risk of mechanical wear and tear, and improving long-term reliability. Moreover, the integration with data acquisition and visualization tools like LabVIEW allows for a higher level of monitoring and control, which is critical in industrial applications where precision is paramount.

However, the system also has some limitations when compared to more advanced model-based control systems, such as those using Model Predictive Control (MPC) or adaptive control techniques.

These advanced systems can handle more complex scenarios and provide better performance in the presence of nonlinearities or time-varying disturbances. Incorporating such techniques into the current system could further improve its accuracy and adaptability.

11.3 Potential for Future Improvements

Several areas for future improvement were identified during the project. First, enhancing the sensor's accuracy at low water levels could be achieved by integrating additional sensors or using alternative measurement techniques, such as capacitive or optical sensors. Second, the development of an automatic calibration algorithm would reduce the need for manual intervention, making the system more user-friendly and adaptable to different environments.

Another potential improvement is the integration of predictive control strategies, such as MPC, which could optimize the system's response to anticipated changes in water inflow or outflow. This would make the system more resilient to disturbances and improve its overall performance. Additionally, expanding the system's capabilities to include remote monitoring and control via IoT platforms would increase its applicability in larger, distributed systems (Ismail, 2019).

11.4 Real-World Applications

The real-time model-based control of a water tank system has numerous applications in industries ranging from chemical processing to water treatment and irrigation. The system developed in this project is particularly suited for scenarios where precise control of fluid levels is critical, such as in chemical reactors, storage tanks, or water distribution networks. Its scalability and adaptability make it a viable solution for both small-scale and large-scale applications.

Moreover, the principles and techniques demonstrated in this project can be applied to other fluid management systems, such as oil reservoirs, fuel tanks, or even biomedical applications like dialysis machines. The integration of real-time data acquisition, control, and monitoring is essential for ensuring the efficiency, safety, and reliability of these systems (Peter-Jules van Overloop, 2006).

12 Conclusion

The project successfully demonstrated the real-time model-based control of a water tank system using a combination of hardware and software tools. The integration of an Arduino-based ultrasonic sensor for water level measurement, coupled with data acquisition and visualization through the NI 6008 board, Simulink, and LabVIEW, provided a robust and accurate control system. The system's ability to maintain the water level within a specified range, despite disturbances, highlights its potential for various industrial applications.

The results and discussion underscore the importance of precise control in fluid management systems and point to potential improvements in sensor accuracy, system calibration, and control algorithms. Future work could focus on integrating predictive control strategies, enhancing sensor capabilities, and expanding the system's applicability to larger and more complex environments.

13 References

American (2012). *Computer modeling of water distribution systems. monograph.* Denver, Co: American Water Works Association.

Amir Reza Neshastehriz (2014). *Robust Model Predictive Control Based Reference Planning in Automated Irrigation Networks*.

Arte Boghossian, James Brown, & Sara Zak (2020). *9.2: P, I, D, PI, PD, and PID control*. [online] Engineering LibreTexts. Available at: https://eng.libretexts.org/Bookshelves/Industrial_and_Systems_Engineering/Chemical_Process_Dynamics_and_Controls_(Woolf)/09%3A_Proportional-Integral-Derivative_(PID)_Control/9.02%3A_P_I_D_PI_PD_and_PID_control [Accessed 27 Aug. 2024].

Arduino (2018). *What Is Arduino?* [online] Arduino water tank Available at: https://www.arduino.cc/en/guide/introduction.

Arduino Forum (2024a). *Arduino Code for water level control*. [online] Arduino Forum. Available at: https://forum.arduino.cc/t/arduino-code-for-water-level-control/1250591 [Accessed 27 Aug. 2024].

Arduino Forum. (2011). *Ultrasonic sensor to determine water level?* [online] Available at: https://forum.arduino.cc/t/ultrasonic-sensor-to-determine-water-level/64890.

Arduino Forum. (2018). *Why accuracy of ultrasonic sensor HC-SR04 so bad for water level?* [online] Available at: https://forum.arduino.cc/t/why-accuracy-of-ultrasonic-sensor-hc-sr04-so-bad-for-water-level/540520 [Accessed 27 Aug. 2024].

Arduino Forum. (2024b). *Water Tank Project - UPDATE*. [online] Available at: https://forum.arduino.cc/t/water-tank-project-update/1207893 [Accessed 27 Aug. 2024].

Bae, Y., Bhattacharya, S., Cui, B., Lee, S., Li, Y., Zhang, L., Im, P., Adetola, V., Vrabie, D., Leach, M. and Kuruganti, T. (2021). Sensor impacts on building and HVAC controls: A critical review for building energy performance. *Advances in Applied Energy*, [online] 4(100068), p.100068. doi:https://doi.org/10.1016/j.adapen.2021.100068.

Chafa, A.T., Chirinda, G.P. and Matope, S. (2022). Design of a real-time water quality monitoring and control system using the Internet of Things (IoT). *Cogent Engineering*, 9(1). doi https://doi.org/10.1080/23311916.2022.2143054.

Chan, K., Schillereff, D.N., Baas, A.C., Chadwick, M.A., Main, B., Mulligan, M., O'Shea, F.T., Pearce, R., Smith, T.E., van Soesbergen, A., Tebbs, E. and Thompson, J. (2020). Low-cost electronic sensors for environmental research: Pitfalls and opportunities. *Progress in Physical Geography: Earth and Environment*, 45(3), pp.305–338. doi:https://doi.org/10.1177/0309133320956567.

Coase, R.H. (2020). The Problem of Social Cost. *The Journal of Law and Economics*, 3(3), pp.1–44. doi:https://doi.org/10.1086/466560.

Dejan (2018). *Ultrasonic Sensor HC-SR04 and Arduino Tutorial*. [online] HowToMechatronics. Available at: https://howtomechatronics.com/tutorials/arduino/ultrasonic-sensor-hc-sr04/.

Dean (2022). *Ultrasonic Sensor HC-SR04 and Arduino - Complete Guide*. [online] How To Mechatronics. Available at: https://howtomechatronics.com/tutorials/arduino/ultrasonic-sensor-hc-sr04/#google_vignette.

Denno, M. (2017). *Strange Results at Tank Outlet with InterFoam -- CFD Online Discussion Forums*. [online] Cfd-online.com. Available at: https://www.cfd-online.com/Forums/openfoam/119571-strange-results-tank-outlet-interfoam.html [Accessed 27 Aug. 2024].

Grabcad.com. (2024). *Free CAD Designs, Files & 3D Models | The GrabCAD Community Library*. [online] Available at: https://grabcad.com/library/tag/hc-sr04 [Accessed 27 Aug. 2024].

Hebert, A.J. (1986). *Analysis of the Feasibility of Using Adaptive, Predictive Control for Heating, Ventilation, and Air Conditioning Systems*.

https://forums.ni.com/t5/user/viewprofilepage/user-id/142275 (2020). *Acquire data using USB6008 in Matlab Simulink*. [online] Ni.com. Available at: https://forums.ni.com/t5/Multifunction-DAQ/Acquire-data-using-usb6008-in-matlab-simulink/td-p/4103542 [Accessed 27 Aug. 2024].

https://forums.ni.com/t5/user/viewprofilepage/user-id/638979 (2020). *Using LabVIEW and a USB-6008 for data acquisition*. [online] Ni.com. Available at: https://forums.ni.com/t5/LabVIEW/Using-LabVIEW-and-a-USB-6008-for-data-acquisition/td-p/4008978.

https://www.facebook.com/realrender3d.co.uk (2023). *How 3D Interior Design Is Demystifying 3D*

Rendering for Interior Designers - RealRender3D. [online] RealRender3D. Available at: https://www.realrender3d.co.uk/how-3d-interior-design-is-demystifying-3d-rendering-for-interior-designers/ [Accessed 27 Aug. 2024].

Ibaseta, D., García, A., Álvarez, M., Garzón, B., Díez, F., Coca, P., Pero, C.D. and Molleda, J. (2021). Monitoring and control of energy consumption in buildings using WoT: A novel approach for smart retrofit. *Sustainable Cities and Society*, 65(102637), p.102637. doi:https://doi.org/10.1016/j.scs.2020.102637.

Ismail Essamlali, Hasna Nhaila and Mohamed El Khaili (2024). Advances in machine learning and IoT for water quality monitoring: A comprehensive review. *Heliyon*, 10(6), pp.e27920–e27920. doi:https://doi.org/10.1016/j.heliyon.2024.e27920.

Ismail, Y. (2019). *Internet of Things (IoT) for Automated and Smart Applications*. BoD – Books on Demand.

Kapoor, R. (2023). *Arduino vs. Other Microcontrollers: Which One is Right for You?* [online] Kunkune. Available at: https://kunkune.co.uk/blog/arduino-vs-other-microcontrollers-which-one-is-right-for-you/?srsltid=AfmBOop009pyG7bsv5B8AeyDx8Z03ZOPNm5Wgx4EBLpcNxyHR9rul_Tj [Accessed 27 Aug. 2024].

Krayem, A. (2024). *Water Level Monitoring: Key Technologies and Applications*. [online] Waltero. Available at: https://waltero.com/water/water-level-monitoring/ [Accessed 17 Jun. 2024].

Last Minute Engineers. (2018). *How HC-SR04 Ultrasonic Sensor Works & How to Interface It with Arduino*. [online] Available at: https://lastminuteengineers.com/arduino-sr04-ultrasonic-sensor-tutorial/#google_vignette.

Liu, Y. (2015). *An Investment Optimization Model for the Firms Pursuing Energy Saving and Carbon Emission Reduction Technology*. water-tank

Macmillan, R.H. (2016). *Introduction to the theory of control in mechanical engineering.* Cambridge Univ Press.

Mathworks.com. (2024). *Verification, Validation, and Test - MATLAB & Simulink Solutions*. [online] Available at: https://uk.mathworks.com/solutions/verification-validation.html [Accessed 27 Aug. 2024].

NCD (National Control Devices, LLC (2024). *Exploring the Power of Ultrasonic Sensors for Water Level Measurement*. [online] Medium. Available at: https://ncdio.medium.com/exploring-the-power-of-ultrasonic-sensors-for-water-level-measurement-31f704c0b897 [Accessed 27 Aug. 2024].

Ni.com. (2024). *USB Data Acquisition (DAQ) Multifunction I/O Devices*. [online] Available at: https://www.ni.com/en/shop/data-acquisition/entry-level-usb-daq.html?cid=PSEA-7013q000001fLK0AAM-CONS-GOGSE_149917762721&utm_keyword=national%20instruments%20usb&gad_source=1&gclid=EAIaIQobChMIwajvrfqTiAMV-o9QBh2yHR2iEAAYASAAEgI9RvD_BwE [Accessed 27 Aug. 2024].

toilet (2012). *Connecting an LCD Screen and an Ultrasonic Distance Sensor to an Arduino*. [online] Instructables. Available at: https://www.instructables.com/Connecting-an-LCD-Screen-and-an-Ultrasonic-Distanc/.

Omar, H.M. (2017). Enhancing automatic control learning through Arduino-based projects. *European Journal of Engineering Education*, 43(5), pp.652–663. doi https://doi.org/10.1080/03043797.2017.1390548.

Paolo Favali, Beranzoli, L. and Angelo De Santis (2016). *SEAFLOOR OBSERVATORIES*. Springer.

Pereira, T.S.R., de Carvalho, T.P., Mendes, T.A. and Formiga, K.T.M. (2022). Evaluation of Water Level in Flowing Channels Using Ultrasonic Sensors. *Sustainability*, 14(9), p.5512. doi https://doi.org/10.3390/su14095512.

Peter-Jules van Overloop (2006). *Model Predictive Control on Open Water Systems*. IOS Press.

PiBorg (n.d.). *Ultrasonic Distance Sensor (HC-SR04)*. [online] PiBorg. Available at: https://www.piborg.org/sensors-1136/hc-sr04.

Prashant Kumar (2013). *Forest Monitoring*. Newnes.

Qiu, Z., Lu, Y. and Qiu, Z. (2022). Review of Ultrasonic Ranging Methods and Their Current Challenges. *Micromachines*, [online] 13(4), p.520. doi:https://doi.org/10.3390/mi13040520.

Quaranta, E., Bejarano, M.D., Comoglio, C., Fuentes-Pérez, J.F., Pérez-Díaz, J.I., Sanz-Ronda, F.J., Schletterer, M., Szabo-Meszaros, M. and Tuhtan, J.A. (2023). Digitalization and real-time control to mitigate environmental impacts along rivers: Focus on artificial barriers, hydropower systems and European priorities. *Science of The Total Environment*, [online] 875(162489), p.162489. water-tank

https://doi.org/10.1016/j.scitotenv.2023.162489.

Ramos, S. (2024). *Mastering Distance Measurement: A Comprehensive Guide to Using Ultrasonic Sensor with LCD and Arduino*. [online] Arduino Intro. Available at: https://arduinointro.com/articles/projects/mastering-distance-measurement-a-comprehensive-guide-to-using-ultrasonic-sensor-with-lcd-and-arduino [Accessed 27 Aug. 2024].

Ranjan, S. (2004). *Control Algorithms for a Two Tank Liquid Level System: An Experimental Study*. [online] Available at: https://core.ac.uk/download/pdf/80147466.pdf.

Singh, B.J., Chakraborty, A. and Sehgal, R. (2023). A systematic review of industrial wastewater management: Evaluating challenges and enablers. *Journal of Environmental Management*, [online] 348(119230), p.119230. doi https://doi.org/10.1016/j.jenvman.2023.119230.

Suman Saurabh, Wankhede Anurag, Singh Siddharth and Yadav Abhiram (2023). Automatic water level indicator and controller. *i-manager's journal on instrumentation and control engineering/i-manager's journal of instrumentation and control engineering*, 11(1), pp.20–20. doi:https://doi.org/10.26634/jic.11.1.20166.

tec-science (2019). *Discharge of liquids (Torricelli's law)*. [online] tec-science. Available at: https://www.tec-science.com/mechanics/gases-and-liquids/discharge-outflow-liquid-speed-torricellis-law/.

uk.mathworks.com. (n.d.). *Continuous-time or discrete-time PID controller - Simulink - MathWorks United Kingdom*. [online] Available at: https://uk.mathworks.com/help/simulink/slref/pidcontroller.html.

Várnai, P., Richards, W.G. and Lyne, P.D. (1999). Modeling the catalytic reaction in human aldose reductase. *Proteins: Structure, Function, and Genetics*, 37(2), pp.218–227. doi:https://doi.org/10.1002/(sici)1097-0134(19991101)37:2%3C218::aid-prot7%3E3.0.co;2-e.

VinayakNairhttps://talkingstuff.net/ (2016). *Video: Water Level Indicator Using an HC-SR04 and an Arduino*. [online] Instructables. Available at: https://www.instructables.com/Ultrasonic-Water-Level-Indicator-Using-HC-SR04-Ard/.

Vojtesek, J., Dostal, P. and Maslan, M. (n.d.). *MODELLING AND SIMULATION OF WATER TANK*. [online] Available at: https://www.scs-

europe.net/dlib/2014/ecms14papers/mct_ECMS2014_0110.pdf [Accessed 27 Aug. 2024].

Wikipedia Contributors (2019). *Ziegler–Nichols method*. [online] Wikipedia. Available at: https://en.wikipedia.org/wiki/Ziegler%E2%80%93Nichols_method.

Zhao, J., Hu, T., Zheng, R., Ba, P., and Zhang, Q. (2021). Design and Performance Analysis of an Ultrasonic System for Health Monitoring of Concrete Structure. *Sensors*, [online] 21(19), pp.6606–6606. doi:https://doi.org/10.3390/s21196606.